

FSUP1024

## Pentane

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 正戊烷  
 Pentane

**Cat No. :** P/1024/17, P/1024/15  
**Synonyms** normal pentane; n-Pentane; Amyl hydride  
**CAS No** 109-66-0  
**Molecular Formula** C5 H12

**Supplier**
**UK entity/business name**  
 Fisher Scientific UK  
 Bishop Meadow Road, Loughborough,  
 Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
 Thermo Fisher Scientific  
 Janssen Pharmaceuticaaan 3a  
 2440 Geel, Belgium

**Emergency Telephone Number** Tel: 01509 231166  
 Chemtrec US: (800) 424-9300  
 Chemtrec EU: 001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
 Liquid

**Appearance**  
 Clear

**Odor**  
 Petroleum distillates

#### Emergency Overview

Highly flammable liquid and vapor. May be fatal if swallowed and enters airways. May cause drowsiness and dizziness. May be harmful in contact with skin. Toxic to aquatic life with long lasting effects. Repeated exposure may cause skin dryness or cracking.

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Flammable liquids.                                 | Category 2 |
| Aspiration Toxicity                                | Category 1 |
| Acute Dermal Toxicity                              | Category 5 |
| Specific target organ toxicity - (single exposure) | Category 3 |
| Acute aquatic toxicity                             | Category 2 |
| Chronic aquatic toxicity                           | Category 2 |

#### Label Elements

## Pentane



## Signal Word

## Danger

## Hazard Statements

H225 - Highly flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H336 - May cause drowsiness or dizziness  
H313 - May be harmful in contact with skin  
H411 - Toxic to aquatic life with long lasting effects

## Precautionary Statements

## Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P271 - Use only outdoors or in a well-ventilated area

## Response

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P331 - Do NOT induce vomiting  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

Vapors may cause flash fire or explosion. Highly flammable.

## Health Hazards

Aspiration hazard if swallowed - can enter lungs and cause damage. May cause drowsiness or dizziness. May be harmful in contact with skin.

## Environmental hazards

Toxic to aquatic life with long lasting effects. Will likely be mobile in the environment due to its volatility. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces.

This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No   | Weight % |
|-----------|----------|----------|
| n-Pentane | 109-66-0 | >95      |

## SECTION 4. FIRST AID MEASURES

## Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.

**Inhalation**

Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention immediately if symptoms occur. Risk of serious damage to the lungs (by aspiration). If not breathing, give artificial respiration.

**Ingestion**

Aspiration hazard. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

**Most important symptoms and effects**

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Dry chemical. Powder. Alcohol resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

**Specific Hazards Arising from the Chemical**

Extremely flammable. Risk of ignition. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

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Use only under a chemical fume hood. Wear personal protective equipment/face protection. Keep away from open flames, hot surfaces and sources of ignition. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Take precautionary measures against static discharges. Avoid contact with skin, eyes or clothing. Do not breathe mist/vapors/spray. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component | China                                                      | Taiwan                                      | Thailand      | Hong Kong                                   |
|-----------|------------------------------------------------------------|---------------------------------------------|---------------|---------------------------------------------|
| n-Pentane | TWA: 500 mg/m <sup>3</sup><br>STEL: 1000 mg/m <sup>3</sup> | TWA: 600 ppm<br>TWA: 1770 mg/m <sup>3</sup> | TWA: 1000 ppm | TWA: 600 ppm<br>TWA: 1770 mg/m <sup>3</sup> |

| Component | ACGIH TLV     | OSHA PEL                                                                                                                                                                             | NIOSH                                                                                                               | The United Kingdom                                                                                                    | European Union                                           |
|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| n-Pentane | TWA: 1000 ppm | (Vacated) TWA: 600 ppm<br>(Vacated) TWA: 1800 mg/m <sup>3</sup><br>(Vacated) STEL: 750 ppm<br>(Vacated) STEL: 2250 mg/m <sup>3</sup><br>TWA: 1000 ppm<br>TWA: 2950 mg/m <sup>3</sup> | IDLH: 1500 ppm<br>TWA: 120 ppm<br>TWA: 350 mg/m <sup>3</sup><br>Ceiling: 610 ppm<br>Ceiling: 1800 mg/m <sup>3</sup> | STEL: 1800 ppm 15 min<br>STEL: 5400 mg/m <sup>3</sup> 15 min<br>TWA: 600 ppm 8 hr<br>TWA: 1800 mg/m <sup>3</sup> 8 hr | TWA: 1000 ppm (8hr)<br>TWA: 3000 mg/m <sup>3</sup> (8hr) |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Viton (R)      | recommendations   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

## Pentane

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                   |                                                                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>   | Wear appropriate protective gloves and clothing to prevent skin exposure                                                                            |
| <b>Respiratory Protection</b>     | No protective equipment is needed under normal use conditions.                                                                                      |
| <b>Large scale/emergency use</b>  | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced |
| <b>Small scale/Laboratory use</b> | Maintain adequate ventilation                                                                                                                       |

|                                        |                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                          |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                            |                                             |
|------------------------------------------------|----------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Clear                      |                                             |
| <b>Physical State</b>                          | Liquid                     |                                             |
| <b>Odor</b>                                    | Petroleum distillates      |                                             |
| <b>Odor Threshold</b>                          | No data available          |                                             |
| <b>pH</b>                                      | No information available   |                                             |
| <b>Melting Point/Range</b>                     | -130 °C / -202 °F          |                                             |
| <b>Softening Point</b>                         | No data available          |                                             |
| <b>Boiling Point/Range</b>                     | 36 °C / 96.8 °F            | @ 760 mmHg                                  |
| <b>Flash Point</b>                             | -49 °C / -56.2 °F          | <b>Method</b> - No information available    |
| <b>Evaporation Rate</b>                        | 28.6 (Butyl Acetate = 1.0) |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable             | Liquid                                      |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.4 vol%      |                                             |
|                                                | <b>Upper</b> 8 vol%        |                                             |
| <b>Vapor Pressure</b>                          | 573 mbar @ 20 °C           |                                             |
| <b>Vapor Density</b>                           | 2.5 (Air = 1.0)            | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.626                      |                                             |
| <b>Bulk Density</b>                            | Not applicable             | Liquid                                      |
| <b>Water Solubility</b>                        | Insoluble                  |                                             |
| <b>Solubility in other solvents</b>            | No information available   |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                            |                                             |
| <b>Component</b>                               | <b>log Pow</b>             |                                             |
| n-Pentane                                      | 3.45                       |                                             |
| <b>Autoignition Temperature</b>                | 260 °C / 500 °F            |                                             |
| <b>Decomposition Temperature</b>               | No data available          |                                             |
| <b>Viscosity</b>                               | 0.25 mPa.s @ 20 °C         |                                             |
| <b>Explosive Properties</b>                    |                            | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available   |                                             |
| <b>Molecular Formula</b>                       | C5 H12                     |                                             |
| <b>Molecular Weight</b>                        | 72.15                      |                                             |

## SECTION 10. STABILITY AND REACTIVITY

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|

## Pentane

|                                         |                                                                                                                   |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                     |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                          |
| <b>Conditions to Avoid</b>              | Incompatible products. Heat, flames and sparks. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Materials to avoid</b>               | Strong oxidizing agents. Halogens.                                                                                |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                                          |

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component | LD50 Oral            | LD50 Dermal           | LC50 Inhalation                  |
|-----------|----------------------|-----------------------|----------------------------------|
| n-Pentane | > 2000 mg/kg ( Rat ) | 3000 mg/kg ( Rabbit ) | 364 g/m <sup>3</sup> ( Rat ) 4 h |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

## (d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs

Central nervous system (CNS)

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs

None known.

(j) aspiration hazard; Category 1

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component | Freshwater Fish | Water Flea | Freshwater Algae | Microtox |
|-----------|-----------------|------------|------------------|----------|
|-----------|-----------------|------------|------------------|----------|

## SAFETY DATA SHEET

## Pentane

|           |                                                                                                                                                           |                                           |  |  |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--|--|
| n-Pentane | LC50: = 9.99 mg/L, 96h<br>(Lepomis macrochirus)<br>LC50: = 11.59 mg/L,<br>96h (Pimephales<br>promelas)<br>LC50: = 9.87 mg/L, 96h<br>(Oncorhynchus mykiss) | EC50: = 9.74 mg/L, 48h<br>(Daphnia magna) |  |  |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--|--|

## Persistence and Degradability

## Persistence

Persistence is unlikely, based on information available.

Degradation in sewage  
treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative Potential

Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| n-Pentane | 3.45    | No data available             |

## Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors.

## Persistent Organic Pollutant

This product does not contain any known or suspected substance.

## Ozone Depletion Potential

This product does not contain any known or suspected substance.

## SECTION 13. DISPOSAL CONSIDERATIONS

Waste from Residues/Unused  
Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

## Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

## Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

## SECTION 14. TRANSPORT INFORMATION

## Road and Rail Transport

|                      |          |
|----------------------|----------|
| UN-No                | UN1265   |
| Proper Shipping Name | PENTANES |
| Hazard Class         | 3        |
| Packing Group        | II       |

## IMDG/IMO

|                      |          |
|----------------------|----------|
| UN-No                | UN1265   |
| Proper Shipping Name | PENTANES |
| Hazard Class         | 3        |
| Packing Group        | II       |

## IATA

## Pentane

UN-No UN1265  
 Proper Shipping Name PENTANES  
 Hazard Class 3  
 Packing Group II

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| n-Pentane | X                                                   | X                                       | X    | X     | 203-692-4 | X    | X   | X     | X    | X    | X    | KE-27968 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Creation Date 14-May-2009  
 Revision Date 04-Apr-2024  
 Revision Summary Not applicable.

## Training Advice

Chemical incident response training.

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

## Key literature references and sources for data



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<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**